

AI StudyBuddy

An Intelligent Personalized Learning Assistant for Students

1. Project Objective

AI StudyBuddy is an intelligent study assistant designed to help students understand study materials, generate notes, practice quizzes, create study plans, and track learning progress using Artificial Intelligence.

2. Main Features

- AI Chat Tutor – Students can ask doubts and receive AI-generated explanations.
- PDF/Notes Upload – Upload study materials and ask questions based on the content.
- AI Notes Generator – Converts lengthy content into concise study notes.
- Quiz Generator – Creates topic-based multiple-choice questions for practice.
- Flashcards – Generates revision flashcards from study materials.
- Study Planner – Creates a study schedule based on subjects and exam dates.
- Progress Tracking – Tracks quiz scores and learning progress.
- Tamil + English Support – Supports bilingual learning.
- Text-to-Speech – Reads AI-generated answers aloud.
- Personalized AI Tutor – Adjusts explanations according to the student's learning level.

3. Suggested Technology Stack

Component	Technology
Frontend	HTML, CSS, JavaScript / React
Backend	Python Flask / FastAPI
Database	MongoDB / MySQL
AI	Gemini API / OpenAI API
PDF Processing	PyMuPDF
Architecture	MVC + REST API + RAG

4. Basic Workflow

Student Login → Dashboard → Upload Material → AI Processing → Ask Question / Generate Notes / Generate Quiz → Save Results → Track Progress

5. Project Title

“AI StudyBuddy – An Intelligent Personalized Learning Assistant for Students”

This document provides the project overview and can be expanded into a complete project report with ER diagram, DFD, UML diagrams, database design, technical architecture, source code, PPT, and documentation.